In order to apply mathematical methods to a physical or "real life" problem, we must formulate the problem in mathematical terms; that is, we must construct a mathematical model for the problem. Many physical problems concern relationships between changing quantities. Since rates of change are represented mathematically by derivatives, mathematical models often involve equations relating to an unknown function and one or more of its derivatives. Such equations are differential equations. They are of basic significance in science and engineering since many physical laws as well as relations are modelled in the form of differential equations. Differential equations are much useful in describing mathematical models involving population growth or radio-active decay. The study of biological sciences and economics is incomplete without the application of differential equations.

The subject of differential equations was invented along with calculus by Newton and Leibniz in order to solve problems in geometry and physics. It played a crucial part in the development of Newtonian physics by the Bernoulli family, Euler, and others. Some of the applications of differential equations in our daily life are found in mobile phones, motor cars, air flights, weather forecast, internet, health care, or in many other daily activities.



Johann Bernoulli
(1667-1748)

In this chapter, we introduce and discuss the first order ordinary differential equations and some methods to find their solutions.

Learning Objectives

Upon completion of this chapter, students will be able to

- classify differential equations
- construct differential equations
- find the order and degree of the differential equations
- solve differential equation using the methods of variables separable, substitution, integrating factor
- apply differential equation in real life problems

10.2 Differential Equation, Order, and Degree

Definition 10.1

A **differential equation** is any equation which contains at least one derivative of an unknown function, either ordinary derivative or partial derivative.

For instance, let $y = f(x)$ where y is a dependent variable (f is an unknown function) and x is an independent variable.

(1) The equation $\frac{dy}{dx} = 0$ is a differential equation.

(2) The equation $\frac{dy}{dx} = \sin x$ is a differential equation.

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